



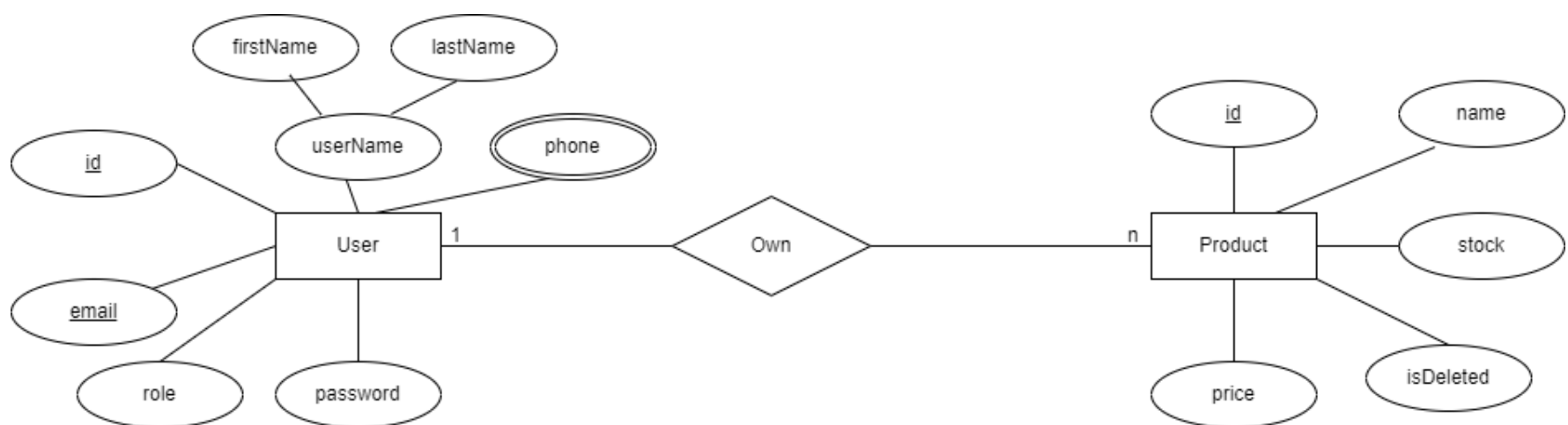
Assignment5

Part 1: ERD Diagram (1 Grade)

Musicana records have decided to store information on musicians who perform on their albums in a database. The company has wisely chosen to hire you as a database designer.

- Each musician that is recorded at Musicana has an ID number, a name, an address (street, city) and a phone number.
- Each instrument that is used in songs recorded at Musicana has a unique name and a musical key (e.g., C, B-flat, E-flat).
- Each album that is recorded at the Musicana label has a unique title, a copyright date, and an album identifier.
- Each song recorded at Musicana has a unique title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A producer may produce several albums.

Part2: Design a schema (Mapping) for the following ERD. (Use any design tool you want) (1 Grade)



Part 3: (Using Node.js and MySQL) Answer the Questions below based on the given Scenario

The small retail store needs a database to manage information about its products, suppliers, and sales.

Database Requirements

1. Products Table:

- **ProductID:** Unique identifier for each product (integer, primary key, auto-increment).
- **ProductName:** Name of the product (**text**).
- **Price:** Price of the product (**decimal**).
- **StockQuantity:** Quantity of the product in stock (**integer**).
- **SupplierID:** ID of the supplier providing the product (**integer, foreign key referencing Suppliers**).

2. Suppliers Table:

- **SupplierID:** Unique identifier for each product (integer, primary key, auto-increment).
- **SupplierName:** Name of the supplier (**text**).



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- **ContactNumber:** Supplier's contact number (**text**).

3. Sales Table:

- **SaleID:** Unique identifier for each product (integer, primary key, auto-increment).
- **ProductID:** Reference to the product sold (**integer, foreign key referencing Products**).
- **QuantitySold:** Quantity of the product sold (**integer**).
- **SaleDate:** Date of sale (**date**).

(Using Node.js and MySQL) generate queries that perform the following tasks (8 Grades):

- 1- Create the required tables for the retail store database based on the tables structure and relationships. (0.5 Grade)
- 2- Add a column "**Category**" to the Products table. (0.5 Grade)
- 3- Remove the "**Category**" column from Products. (0.5 Grade)
- 4- Change "**ContactNumber**" column in Suppliers to **VARCHAR (15)**. (0.5 Grade)
- 5- Add a **NOT NULL** constraint to ProductName. (0.5 Grade)
- 6- **Perform Basic Inserts:** (0.5 Grade)
 - a. Add a supplier with the name '**FreshFoods**' and contact number '**01001234567**'.
 - b. Insert the following three products, all provided by '**FreshFoods**':
 - i. '**Milk**' with a price of **15.00** and stock quantity of **50**.
 - ii. '**Bread**' with a price of **10.00** and stock quantity of **30**.
 - iii. '**Eggs**' with a price of **20.00** and stock quantity of **40**.
 - c. Add a record for the sale of **2** units of '**Milk**' made on '**2025-05-20**'.
- 7- Update the price of '**Bread**' to **25.00**. (0.5 Grade)
- 8- Delete the product '**Eggs**'. (0.5 Grade)
- 9- **Retrieve the total quantity sold for each product.** (0.5 Grade)
- 10- Get the product with the highest stock. (0.5 Grade)
- 11- Find suppliers with names starting with 'F'. (0.5 Grade)
- 12- Show all products that have never been sold. (0.5 Grade)
- 13- Get all sales along with product name and sale date. (0.5 Grade)
- 14- Create a user "**store_manager**" and give them **SELECT, INSERT, and UPDATE** permissions on **all tables**. (0.5 Grade)
- 15- Revoke **UPDATE** permission from "**store_manager**". (0.5 Grade)
- 16- Grant **DELETE** permission to "**store_manager**" only on the Sales table. (0.5 Grade)

Bonus (2 Grades)

How to deliver the bonus?

- 1- Solve the problem [Customer Who Visited but Did Not Make Any Transactions](#) on **LeetCode**
- 2- Inside your assignment folder, create a **SEPARATE FILE** and name it "bonus.txt"
- 3- Copy the code that you have submitted on the website inside "bonus.txt" file